Topic Modeling and Semantic Mapping of French Political Manifestos

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Abstract

In this project, we use the Archelec corpus of French legislative *professions de foi* from 1973 to 1993 (21,156 documents) to examine whether political families from this period can be mapped within a shared semantic space using text alone. Our approach combines OCR quality filtering, a preprocessing pipeline, transformer-based embeddings computed on overlapping 150-word chunks, and BERTopic with manually interpreted topics. We find that political families are only partially separable in the embedding space. Thematic specialisation is mainly observed among more peripheral parties (such as Front National, Lutte Ouvrière, and ecologists), while mainstream parties tend to rely on a more generic electoral discourse. Focusing on topics shared across parties, sentiment analysis reveals noticeable differences in tone; the radical left consistently stands out as the most critical. Finally, we observe that repeated text segments within topics point to an organisational pattern: some peripheral parties reuse centrally produced templates that local candidates adopt almost verbatim. Code is available in our project repository.

1 Introduction

Political manifestos are written under fairly strict constraints: each candidate produces one document per round, in a fixed format and addressed directly to voters. Unlike speeches or social media posts, this makes them easy to compare across candidates and elections.

We work on the Archelec corpus, restricted to five legislative elections (1973, 1978, 1981, 1988, 1993), a window that spans the end of the Gaullist era, the 1981 alternation, the Mitterrand–Chirac cohabitation, and the more fragmented landscape of the early 1990s. The central question is whether the political families of this period can be cartographed in a shared semantic space inferred from manifesto content alone. The shared part of the project, developed with a co-author[1], checks whether the induced topics are substantively meaningful, whether thematic profiles cluster by political family, and whether some families develop distinctive signatures. The individual contribution of the present author adds a fourth question: when several families address the same topic, do they frame it with the same tone, or does statistical co-presence mask substantive divergence?

The pipeline addresses three difficulties. OCR noise varies across elections, manifestos mix several themes and resist document-level representations, and a generic electoral vocabulary tends to dominate the topics. We respond with an OCR quality filter, a two-track preprocessing scheme, 150-word overlapping chunks, and a BERTopic configuration tuned on this corpus. A sentiment analysis on shared topics extends the reading to inter-family tonal divergence.

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2 Related Work

Political manifesto analysis has traditionally relied on expert coding. The Comparative Manifesto Project is the main reference: it assigns manifesto content to predefined policy categories and enables cross-national comparison (Volkens et al., 2013). This approach, however, is time-consuming, costly, and difficult to scale to large or less standardised corpora. Quantitative alternatives such as Wordscores and Wordfish (Slapin and Proksch, 2008) estimate ideological positions from word frequencies, but their focus on a dominant ideological dimension is less suited to fragmented party systems where multiple thematic cleavages coexist.

Latent Dirichlet Allocation (LDA) (Blei et al., 2003) remains the standard probabilistic model for topic extraction, representing documents as mixtures of topics and topics as distributions over words. Its simplicity explains its wide use, but it relies on bag-of-words representations, is sensitive to preprocessing choices, and requires fixing the number of topics in advance. Embedding-based approaches offer an alternative by representing text segments in a continuous semantic space rather than relying on word counts (Reimers and Gurevych, 2019). BERTopic (Grootendorst, 2022) builds on this idea by combining sentence embeddings, UMAP dimensionality reduction (McInnes et al., 2018), HDBSCAN clustering (Campello et al., 2013), and class-based TF-IDF topic representations. This framework is well suited to political texts: it does not require fixing the number of topics a priori, captures semantic similarities beyond surface forms, and allows low-density documents to be treated as noise rather than forced into a topic.

Embedding-based methods have also encouraged a cartographic reading of political language, where groups can be represented as regions in a shared semantic space and distances interpreted as proxies for ideological proximity. This interpretation, however, should be treated with caution: observed structures may reflect preprocessing choices, model architecture, or training data biases as much as substantive differences. While topic modeling has been applied to a wide range of political corpora, French legislative manifestos remain relatively underexplored. Our project applies a tuned BERTopic configuration and validates the resulting structure through quantitative analysis, manual topic interpretation, and qualitative inspection.

3 Data

3.1 Corpus and political families

The analysis uses the Archelec corpus, maintained by CEVIPOF at Sciences Po Paris, which contains OCR-processed *professions de foi* from elections under the French Fifth Republic, together with candidate-level metadata (identifier, date, election, party support, list, profession). Each document corresponds to one electoral tract distributed before the first or second round of a legislative election. We restrict the study to the legislative elections of 1973, 1978, 1981, 1988, and 1993.

The affiliation fields `titulaire-soutien` and `titulaire-liste` contain more than 1,800 distinct labels, including party names, coalitions, local lists, and many vague or missing entries. We group these into seven political families using a set of substring and regular-expression rules, built through manual inspection of the data. To do this, we first examine label frequencies by election year, since patterns change over time: gaullist labels shift from UDR to RPR after 1973, the Front National becomes more visible in later elections, the PSU disappears after the 1970s, and ecologist labels increase sharply by 1993. We also give priority to specific expressions over more general ones (for example, *parti socialiste unifié* before *socialiste*, or *ligue communiste* before *communiste*) to avoid incorrect matches. Remaining unmatched labels were reviewed manually, and the rules were extended to cover coalition names, smaller platforms, and ecologist subgroups. Truly marginal cases were left in *other*. The seven families are: *national right* (FN, Parti des Forces Nouvelles); *radical left* (LO, LCR, PSU, Comités Juquin, POE); *communist left* (PCF); *socialist left* (PS, MRG, Mouvement des Citoyens); *ecologist* (Verts, Génération Écologie); *gaullist right* (UDR, RPR and related labels); and *liberal center right* (UDF, CDS, Républicains Indépendants, Démocratie chrétienne, CNIP). The share of *unclassified* and *other* documents ranges from 13% in 1973 to 41% in 1988. This higher value is mainly due to missing metadata and broad coalition labels, which makes interpretation by political family less reliable for that year.

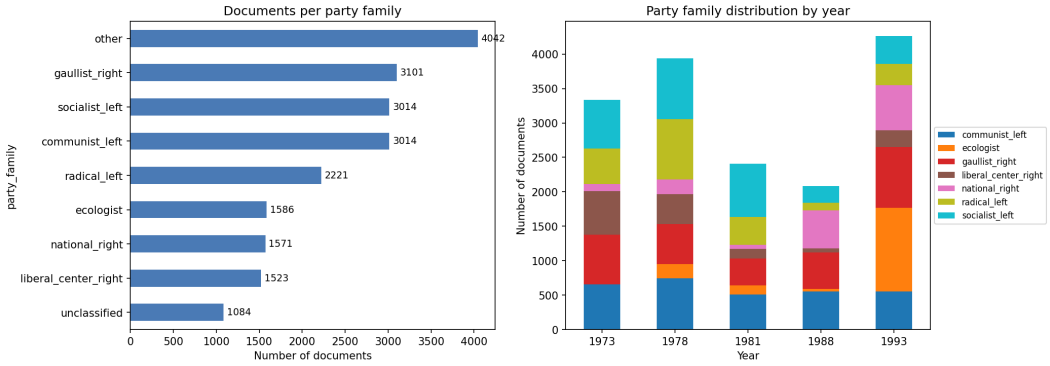

Figure 1: Distribution of documents across political families (left) and family composition by election year (right).

3.2 OCR quality

The texts come from OCR applied to archival electoral documents, so their quality is uneven. Some are clean, while others contain typical OCR errors such as misread characters, broken or merged words, or noise from the original layout. To get a rough idea of text quality, we compute a simple score at the document level based on two elements: the proportion of words that match common French vocabulary (as a proxy for readability), and the share of very short tokens (length ≤ 2), which often indicate segmentation issues. The score ranges from 0 to 1. We keep documents with more than 50 tokens and a limited proportion of short tokens (< 0.5), which allows us to filter out only the most degraded texts. As shown in Figure 2, the average quality remains stable across election years (around 0.42–0.43), with more variation within each year than between years.

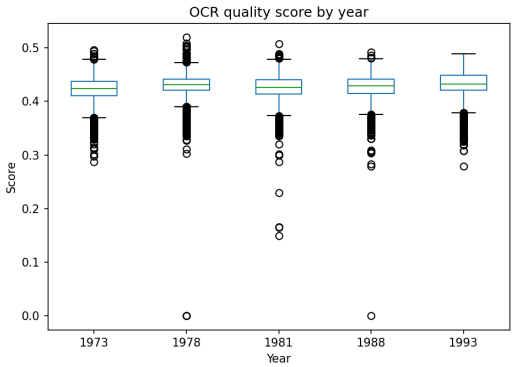

Figure 2: Distribution of OCR quality scores across election years.

3.3 Preprocessing, segmentation, and embeddings

We work with two versions of each document. BERTopic's encoder uses raw text, while its c-TF-IDF step relies on a bag-of-words representation and therefore requires cleaner input. This is why we separate the pipeline. The first version of cleaning is used for embeddings. We only remove clear OCR artefacts (such as institutional headers or imprint blocks), generic electoral expressions (*votez pour*, *chers électeurs*), and isolated single-character tokens (except French clitics). Apart from that, the text is kept as close as possible to its original form.

The second version of preprocessing is used for c-TF-IDF. Starting from the output of the first version, we lemmatise the text with spaCy and then remove around 700 stopwords. The stopword list is adapted to the political context: words such as *gauche*, *droite*, *état*, *immigration*, and *travail* are deliberately kept. On average, this reduces the number of tokens by about 54%.

Most manifestos are longer than the encoder's 512-token limit. Truncating them would remove important sections, often where policy proposals are developed. To avoid this, we split each document into overlapping chunks of 150 words, with a 30-word overlap, and discard very short fragments (less than 40 words). We also tested sentence- and paragraph-based segmentation on a sample of 200 documents, but these approaches performed poorly after preprocessing, as much of the punctuation had been removed. The fixed-length approach gave better results in terms of coherence and diversity.

The final dataset contains about 118,000 chunks, each linked to the metadata of its source document (Figure 3). It is worth noting here that for the family-level analysis in Section 6, we restrict the data to chunks belonging to the seven main political families, excluding those labelled as *other* or *unclassified*. Each chunk is encoded using `paraphrase-multilingual-MiniLM-L12-v2` (Reimers and Gurevych, 2019), which produces 384-dimensional embeddings. These embeddings are L2-normalised and computed once, then reused for topic modeling, clustering, and sentiment analysis.

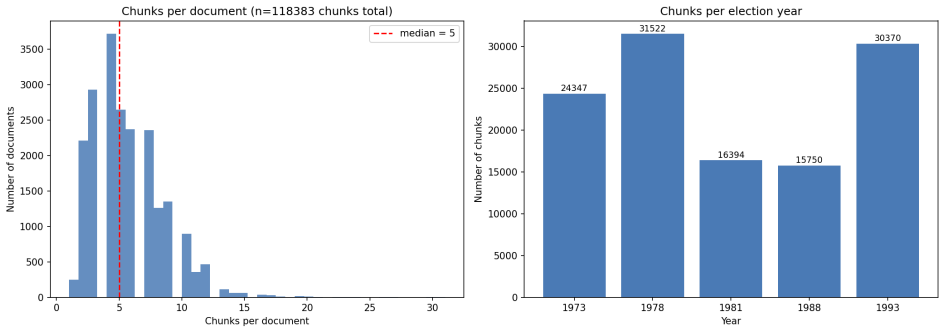

Figure 3: Distribution of chunks per document and per election year after segmentation.

4 Method

The thematic structure of the corpus is extracted with BERTopic (Grootendorst, 2022), an embedding-based topic modeling approach. Electoral manifestos often discuss the same issues using different words. A paragraph about employment may refer to *emploi*, *chômage*, or *licenciement*, which limits methods based purely on word co-occurrence such as LDA (Blei et al., 2003). Methods relying on semantic similarity capture these variations more robustly and tolerate OCR noise better. BERTopic also relies on HDBSCAN for clustering, which does not force every chunk into a topic: low-density regions are treated as noise. This matters here, since many mainstream manifestos share a broad and fairly generic discourse that does not form clearly separated thematic clusters. Finally, BERTopic produces interpretable topic representations through c-TF-IDF, which makes manual inspection and labelling possible. We use unsupervised topic modeling rather than supervised alternatives such as Labeled LDA (Ramage et al., 2009) or seeded LDA (Watanabe and Baturo, 2024), even though party labels are available. The goal is to test whether thematic structure emerges from the text alone and aligns with political families; conditioning the model on labels at training time would make the analyses circular.

4.1 Embedding-based topic modeling: BERTopic

BERTopic operates on chunk-level embeddings. Let $\{c_i\}_{i=1}^{N}$ denote the set of chunks. Each chunk is mapped to a vector representation $\mathbf{e}_i = f(c_i)$, where f is the pretrained transformer-based sentence encoder (Vaswani et al., 2017) and $\mathbf{e}_i \in \mathbb{R}^{384}$. Distances in this space reflect contextual semantic similarity rather than surface word overlap. We use a multilingual encoder rather than a French-specific sentence-CamemBERT variant for its favourable size/quality trade-off; a domain-adapted encoder remains a natural extension.

Dimensionality reduction. Embeddings are projected to a lower-dimensional space using UMAP (McInnes et al., 2018): $\mathbf{z}_i = \mathrm{UMAP}(\mathbf{e}_i)$. The projection preserves local neighbourhoods while making the space easier to cluster. We use cosine distance with $n_{\text{neighbors}} = 30$, $n_{\text{components}} = 5$, and $d_{\min} = 0$ (Section 5).

Density-based clustering. The reduced representations $\mathbf{z}_i$ are clustered with HDBSCAN (Campello et al., 2013), which identifies dense regions without requiring a predefined number of clusters. Observations outside any dense region are labelled as noise (cluster -1). This property matters on our corpus, where a large fraction of the discourse is generic.

Topic representation via c-TF-IDF. Each cluster is interpreted as a topic. To characterise topic k, all its chunks are concatenated into a pseudo-document, and a class-based TF-IDF score is computed:

$$\text{c-TF-IDF}(t,k) = \frac{\text{tf}(t,k)}{\sum_{t'} \text{tf}(t',k)} \cdot \log\left(\frac{N_T}{\text{df}(t)}\right)$$

where $\text{tf}(t,k)$ is the frequency of term t in topic k, $\text{df}(t)$ its frequency across all topic pseudo-documents, and N_T the number of topics. This representation highlights terms that are both frequent within a topic and relatively rare elsewhere.

Noise handling and topic reduction. HDBSCAN labels a substantial number of chunks as noise. BERTopic provides a procedure to reassign these chunks to the nearest topic, but on our corpus this inflates the largest cluster to over 90% (Section 5). We therefore keep noise as a separate category and exclude it from downstream aggregation. The remaining clusters are reduced to $K = 30$ using `reduce_topics` to obtain a taxonomy that can be manually labelled.

4.2 From chunk-level topics to candidate profiles

BERTopic assigns a topic label z_j to each chunk. To analyse documents at the candidate level, we aggregate these labels within each document. For a document d with n_d chunks, its topic profile is the empirical distribution of topics:

$$P(k \mid d) = \frac{1}{n_d^*} \sum_{j=1}^{n_d} \mathbf{1}[z_j = k]\,,$$

where n_d^* excludes noise chunks ($z_j = -1$). This captures the relative importance of each topic in the document. We also compute a document-level embedding by averaging the embeddings of its chunks, which provides a continuous representation of the text independent of the topic assignment.

These representations are used to study the structure of the corpus. First, we inspect the embedding space directly through a UMAP projection to assess whether political families occupy distinct regions. We then analyse how topics are distributed across families by counting their occurrences, which yields a family-by-topic contingency table. This allows us to measure thematic specialisation using the lift

$$\text{lift}(\mathcal{F}, k) = \frac{P(k \mid \mathcal{F})}{P(k)},$$

and to evaluate the overall association between families and topics with a Pearson χ^2 test and Cramér's V.

Finally, we study tonal differences across families. Each chunk is assigned a sentiment score using a French five-star classifier (`cmarkea/distilcamembert-base-sentiment`), rescaled to $s(c) \in [-1, 1]$. On topics shared by several families with sufficient data, we compare average sentiment across families to measure tonal divergence.

5 Experimental setup

This section presents how we tune BERTopic and how we validate the topics without ground-truth labels.

5.1 Hyperparameter selection for BERTopic

BERTopic involves choices at three stages: UMAP projection, HDBSCAN clustering, and c-TF-IDF vectorisation. With default parameters, most chunks fall into a single large cluster, which is not very informative.

We therefore tune the model through qualitative inspection, following (Morin et al., 2026). Concretely, we look at the 2D embedding map, the top words of each topic, and a few representative chunks. We avoid relying only on automated coherence scores, as these can be unstable in unsupervised settings (Stammbach et al., 2023). We start from a standard configuration ($n_{\text{neighbors}} = 50$, `min_cluster_size` $= 50$) and adjust UMAP and HDBSCAN jointly. Smaller values produce many small, noisy clusters, while larger ones merge distinct themes. We retain $n_{\text{neighbors}} = 30$ and `min_cluster_size` $= 60$ as a good compromise.

Choosing the number of topics K and handling noise. We then compare four configurations, combining $K \in \{20, 30\}$ with or without outlier reassignment. Table 1 summarises the results. The table should be read as follows: the "Largest topic" column indicates how dominant the biggest cluster is (a very large value means the model collapses everything into one topic), while the "Noise" column shows the share of chunks that HDBSCAN leaves unassigned. The last columns indicate whether specific political families (PCF, FN, ecologists, LO) are captured by distinct topics or not.

Table 1: Effect of `nr_topics` and `reduce_outliers` on the BERTopic taxonomy.

Configuration	Largest topic	Noise (-1)	PCF	FN	Écolo	LO
$K = 20$, `reduce_outliers`=on	90.3%	0%	–	mix	mix	several
$K = 30$, `reduce_outliers`=on	39.5%	0%	–	several	mix	several
$K = 20$, `reduce_outliers`=off	20.2%	50.5%	fused with LO	–	–	several
$K = 30$, **`reduce_outliers`=off**	$\mathbf{14.4}\%$	$\mathbf{54.3}\%$	✓	✓	✓	✓

Reassigning all noise points tends to inflate the largest cluster (up to 90%), mixing together very different types of discourse. Keeping noise as such leads to more meaningful results. With $K = 20$, the model remains too coarse: for example, PCF and Lutte Ouvrière are grouped into the same topic. With $K = 30$, all major peripheral families are captured by distinct topics, and even smaller parties appear as separate clusters. We therefore retain $K = 30$ without outlier reassignment and exclude noise chunks (about 54%) from downstream analysis. The final configuration uses UMAP ($n_{\text{neighbors}} = 30$), HDBSCAN (`min_cluster_size` $= 60$), and a c-TF-IDF with adapted stopwords and bigrams. Lowering `max_df` prevents generic electoral terms (*candidat*, *circonscription*) from dominating the topics, while bigrams help preserve expressions such as *front national*. This results in a 30-topic taxonomy (including one boilerplate topic) that remains interpretable.

5.2 Validation of topic coherence

In the absence of a reference annotation, which is a common limitation in topic modeling, we assess topic quality using a simple topic digest. For each topic, we examine the top c-TF-IDF terms and three representative chunks, along with three automatic flags: *generic boilerplate*, *repetitive*, and *duplicate chunks*. We sometimes observe duplicate flags. These are not modelling errors but reflect how the corpus is produced: some organisations distribute centrally written campaign texts that local candidates reuse almost word for word.

Manual labelling is carried out in a dedicated notebook[2], where, for each of the 29 substantive topics, we display the top eight c-TF-IDF terms along with five chunks from different documents. In most cases, the political family can be identified directly from the text, through phrases such as *"présenté par lutte ouvrière"* or *"le front national appelle aux français"*. To help structure the analysis and identify a few less well-known organisations (Parti des Forces Nouvelles, Parti Ouvrier Européen, Parti de la Loi Naturelle), we used a large language model as a support tool, following Morin et al. (2026). Its suggestions were used as guidance only and systematically checked against the original chunks before validation.

6 Results

The main question is whether French political families between 1973 and 1993 can be mapped in a shared semantic space using only the content of their manifestos. The seven families used below are those built in Section 3.1. Three questions organise the analysis: are the induced topics readable

[2]The code and analysis are available in the repository.

as political registers, do candidates from the same family cluster together, and are some families more thematically distinctive than others? A final analysis, contributed by myself, asks whether families that share a topic also share the same tone. All results use the final BERTopic configuration of Section 5.

Are the induced topics politically interpretable ? Topic interpretability is checked through the labelling notebook described in Section 5: for each of the 29 substantive BERTopic topics, the top eight c-TF-IDF terms and five representative chunks were inspected, and the corresponding political family was attributed manually. Most topics are recognisable as a clear register or sub-register: extreme-left labour rhetoric, far-right immigration discourse, ecologist platform, socialist Union de la Gauche language, PCF anti-Mitterrand register, and generic electoral boilerplate. Figure 4 shows the top-word distributions of the most distinctive topics. A residual limit affects the c-TF-IDF representation: OCR errors that survive the quality filter occasionally produce odd bigrams in the top-word lists, particularly for the earliest elections.

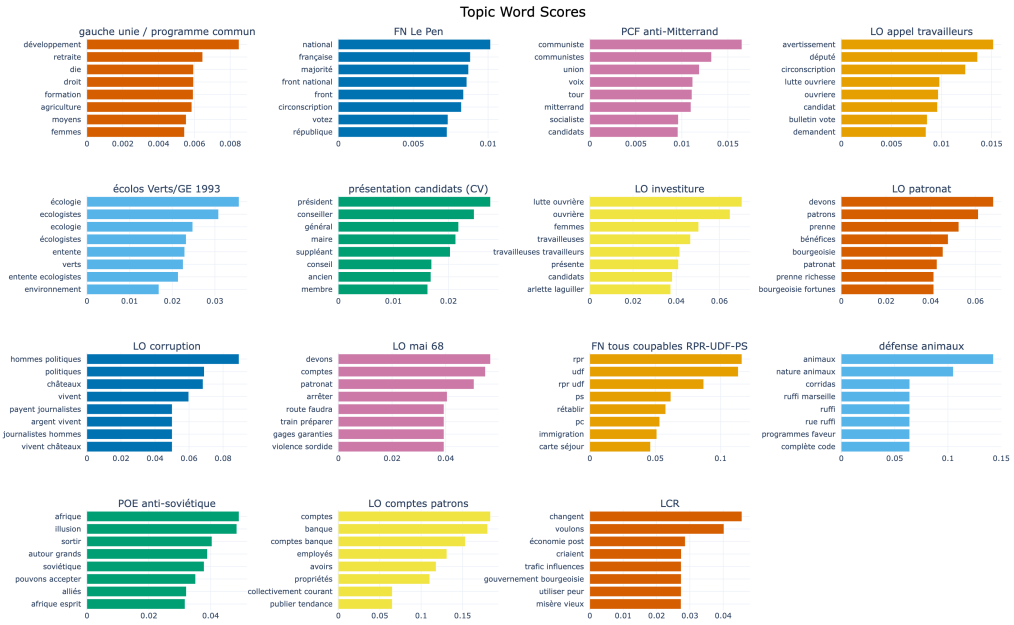

Figure 4: Top-eight c-TF-IDF terms of the most distinctive BERTopic topics, manually labelled.

Do thematic profiles cluster by political family? Figure 5 helps illustrate this result. In the embedding space, peripheral families such as *radical_left*, *ecologist*, and *national_right* form more distinct regions, while mainstream parties (PS, RPR, UDF) are much more overlapping. The right panel further shows that even clearly distinct families (e.g. *national_right* vs *ecologist*) occupy separate areas, confirming that the structure is not random.

Overall, political families are only partially separable: some groups stand out clearly, but a large part of the corpus shares a common discursive space. This confirms the idea of a shared electoral core, with more distinctive positions at the periphery.

Are some families thematically more distinctive than others? The previous results suggest that political families partly overlap in the semantic space, but do not completely coincide. To understand where the differences come from, we compare how topics are distributed across families. Concretely, we count how often each family appears in each topic, which gives a 7×29 table.

A standard χ^2 test shows that this distribution is far from random ($p \ll 10^{-300}$). To measure how strong this association is, we use Cramér's V, which ranges from 0 (no link between family and topic) to 1 (perfect separation). Here, $V = 0.45$, which indicates a fairly strong relationship: political

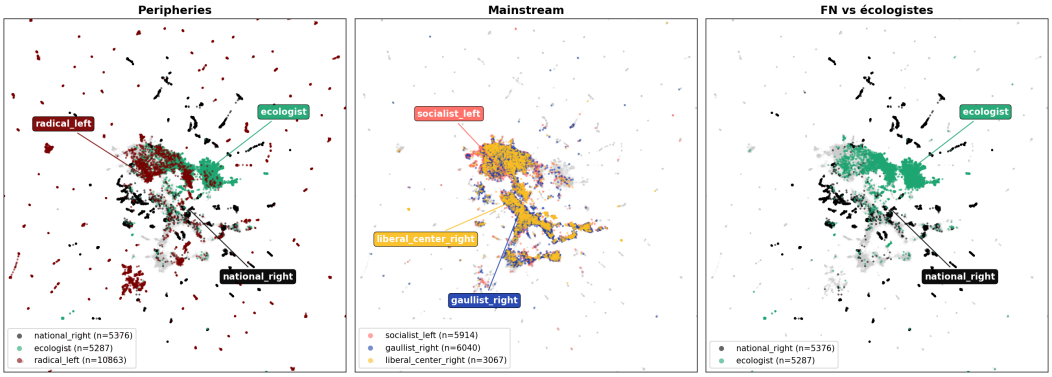

Figure 5: UMAP projection of the 43,980 non-noise chunks of the seven main families in the embedding space.

families are not randomly spread across topics. To go further, we look at where each family is over- or under-represented. For this, we use a simple measure called *lift*: a value above 1 means that a family talks about a topic more than average, while a value below 1 means the opposite.

Figure 6 summarises these patterns. Red cells indicate topics that are over-represented for a given family, while blue cells show under-representation. The contrast is quite clear: peripheral families (*radical_left*, *national_right*, *ecologist*) have strong, concentrated red areas, meaning they focus on a few distinctive topics. In contrast, mainstream parties (PS, RPR, UDF) show a more diffuse pattern, with weaker contrasts across topics.

This confirms an asymmetric structure: peripheral families tend to have clear thematic identities, while mainstream parties share a broader and more generic electoral language. This is consistent with the idea of a common "centre" of political discourse, surrounded by more specialised positions.

Two caveats should be kept in mind. First, about 54% of chunks are excluded as HDBSCAN noise, and these are mostly from mainstream parties. This may partly explain why their profiles appear less distinctive. Second, the 1988 election includes many *unclassified* or *other* documents, which makes the results for that year less reliable (Section 3).

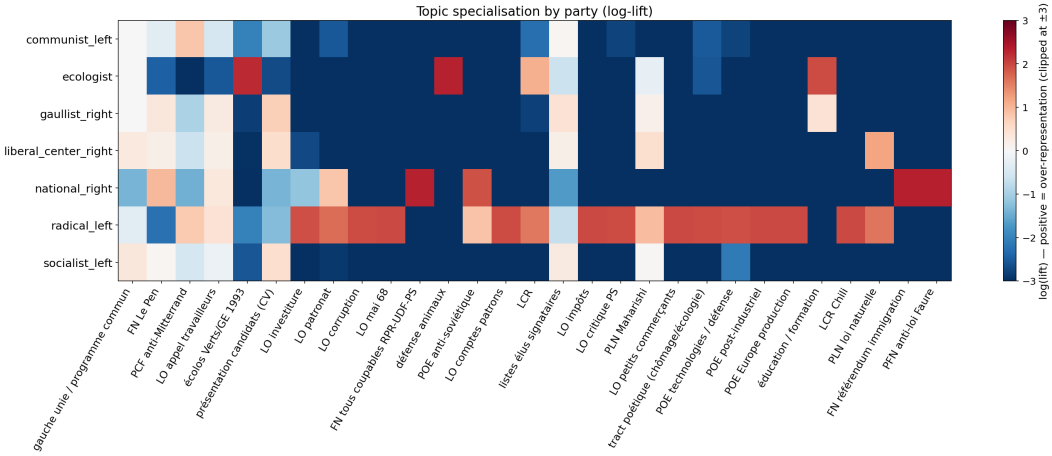

Figure 6: Log-lift of each topic by political family (clipped at ± 3).

When families share a topic, do they frame it the same way ? So far, we have identified topics that are shared across several political families. But talking about the same topic does not mean talking about it in the same way. Different parties may frame the same issue with very different tones. To capture this, we assign a sentiment score to each chunk using a five-level French sentiment classifier introduced in Section 4, and rescale it to a value between -1 and 1. We focus only on topics

that are genuinely shared (at least three families, with enough data for each), and compare the average sentiment of each family within those topics.One important limitation is that the classifier is biased toward positive scores. Because it was trained on product reviews, it tends to interpret the generally promotional tone of manifestos as positive. As a result, most chunks receive a score above zero (with a corpus average of $+0.63$). We therefore interpret sentiment values in relative terms so that what matters is the difference between families, not the absolute level. Despite this bias, clear differences appear across families (Figure 7). The boxplots show consistent shifts in median sentiment across political families within the same topic. On the most contrasted shared topics, the gap reaches around 0.2 to 0.25. While modest in absolute terms, this difference is meaningful given the overall positive baseline. The radical left (LO, LCR, POE) consistently stands out as the most critical voice, while other families tend to adopt a more positive tone.

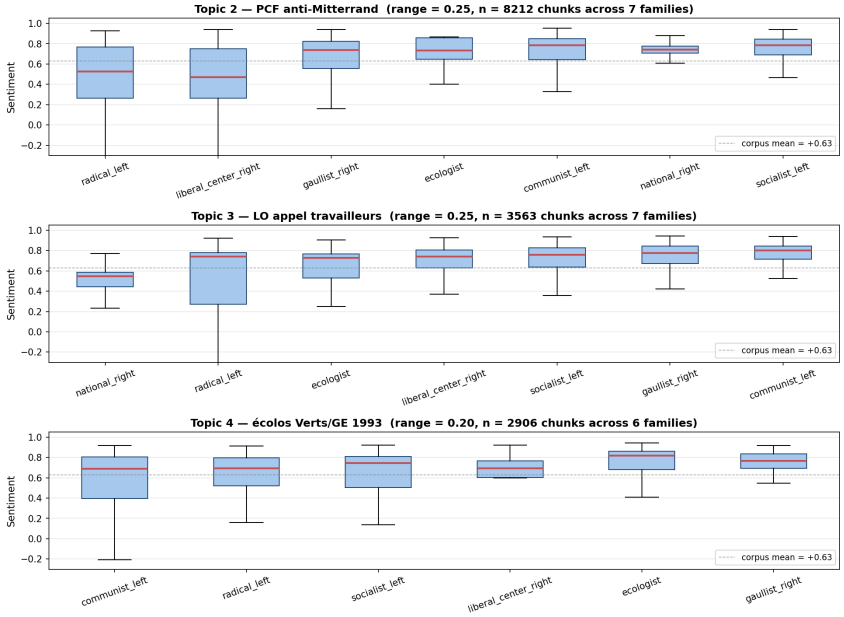

Figure 7: Distribution of sentiment by political family on the most contrasted shared topics. The corpus average is $+0.63$. Families are ordered from most critical (left) to most positive (right).

A closer look at individual examples confirms this pattern. On the same topic, one party may adopt a critical stance while another presents a more supportive or unifying view. In other words, sharing a topic does not imply sharing a perspective: even within the same thematic space, political families can frame issues quite differently.

7 Discussion and Conclusion

Discussion. The results point to an asymmetry in the semantic space. Mainstream parties (PS, RPR, UDF) mostly rely on a broad and rather generic vocabulary, and a large share of their chunks ends up in HDBSCAN noise. When they do stand out, their topics remain only moderately specific, with lifts typically between 1.5 and 3. Peripheral organisations, by contrast (Front National, Lutte Ouvrière, ecologists), show much sharper thematic concentration, with lifts reaching 7 to 10 on their most characteristic topics. The PCF lies somewhere in between, around 2.3. Taken together, this confirms that political families are only partially separable in the embedding space : there is a shared electoral core, and more distinctive positions at the margins. Looking at shared topics gives a slightly different perspective. Even when parties talk about the same issue, they do not necessarily do so in the same way. On the four most contrasted topics, average sentiment differs by as much as 0.25 across families (on the $[-1, 1]$ scale). The radical left is the most critical on two of them (*gauche unie / programme commun* and *PCF anti-Mitterrand*), while the national right and the communist left take that role on the other two. Given the strong positive bias of the classifier (mean $+0.63$, and 95% of chunks above zero), these values are better read as relative differences than as absolute levels. What matters here

is that, within a same topic, parties tend to shift the tone in a consistent way. Another point worth noting is the presence of repeated text. In some peripheral parties, especially Lutte Ouvrière, many candidates reuse centrally written material almost word for word. Mainstream parties, on the other hand, tend to produce more individualised documents. This is not just a technical detail: it affects the structure of the topics, and also reflects differences in how campaigns are organised.

Limitations. Several limitations should be kept in mind. OCR quality is uneven, particularly in earlier elections, and some noise remains in the data. The grouping into seven political families is also a simplification of a more complex landscape. In 1988 in particular, a large fraction of documents remains *unclassified* or *other* (around 40%), which weakens the interpretation for that year. The HDBSCAN noise rate is high (about 54%), which is partly informative but also means that the analysis focuses on a subset of the corpus. Finally, the sentiment analysis should be interpreted with caution: the model is trained on review data rather than political texts, and no formal statistical testing is provided. Simple additions such as bootstrap confidence intervals or non-parametric tests would make the results more robust, and a domain-specific model would likely improve the measurement.

Conclusion. This study combines adapted preprocessing, transformer embeddings, and a tuned BERTopic model to analyse $21{,}156$ French legislative manifestos from 1973 to 1993. The main result is the presence of a dual structure: mainstream parties share a relatively common electoral language, while peripheral parties are more clearly associated with specific themes. The analysis of tone adds another layer, showing that even when topics overlap, the way they are framed can differ across families. Several extensions would be worth exploring, in particular using longer-context models to avoid chunking, and developing sentiment tools better suited to political texts. Extending the corpus to more recent elections would also help assess whether these patterns persist in a different political context.

Contribution Statement

This work was carried out as part of the NLP course at ENSAE Paris (2025–2026), taught by C. Kermorvant. The pipeline, topic validation, clustering, and thematic analysis (Sections 2 to 6) were developed jointly with Clément Destouesse (`clement.destouesse@ensae.fr`). The sentiment analysis on shared topics (Section 6) is my own contribution. The implementation relies on standard documentation and was partly informed by the BERTopic tutorial by A. Morin, É. Schultz, and É. Ollion.

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